STRATEGY REPLICATION · VOLATILITY RISK PREMIUM

The VIX term-structure sleeve

Short volatility in calm markets, long in stressed ones, sized off the VIX level. A complete, self-contained replication of the VIX strategy published by **Predicting Alpha** as the "North Star Portfolio" — rules, arithmetic, results, and both ways to trade it: the ETN/ETF wrapper and VIX futures directly.

10.6%	12.9%	0.85	-20.8%
CAGR	VOLATILITY	SHARPE	MAX DRAWDOWN
2011-2026, VIXY version	vs 17.0% for SPY	daily, 0% risk-free	inside 2020

SOURCE

Predicting Alpha — **North Star Portfolio, "Full Research + Evidence"**
predictingalpha.com/research
Published research window: 2011-01-04 → 2025-10-17
This document: replication as of October 2026, data to 2026-10-22

SCOPE

The **VIX sleeve only**. The All-Weather ETF sleeve and the combined 140%-gross portfolio are deliberately excluded — they are covered in the project's main report.
Both wrappers: ETN/ETF (VIXY, VXX) and VIX futures (front month, monthly roll).

ATTRIBUTION

The strategy is **not ours**. Every rule on the following pages is Predicting Alpha's, restated so that a reader can re-implement it from scratch. The measurement, the proxy choices and the two-wrapper comparison are the work of this replication. The source calls several implementation details proprietary; where that happens the document says so and shows what was chosen instead.

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All results are historical simulations on free daily data with explicit costs. They are not a forecast, not investment advice, and not a track record.

1 · What the strategy is

The VIX futures curve is usually upward sloping: the front contract trades below the second month, which trades below the third. Hold a short position and the futures price drifts down toward spot as expiry approaches — the **roll-down yield**. Investors pay that premium for insurance, and the seller of insurance collects it. That works until volatility spikes, at which point the same position loses several years of carry in days. The strategy is a rule for **when** to be short, **when to be long**, and **how big**.

TWO SIGNALS

1. Volatility risk premium (VRP). Implied volatility versus what recent realised volatility says it should be. When implied is rich relative to realised, protection is overpriced and short volatility is paid. When implied is cheap, shorting is dangerous: the market is not paying you enough for the risk.

2. Term structure. Short-term fear versus medium-term fear. If the 30-day VIX sits below the 3-month VIX3M the curve is in **contango** — a normal, calm market. If the 30-day spikes above it, the curve is in **backwardation** and near-term risk is elevated.

THE DECISION TABLE

VRP	TERM STRUCTURE	POSITION
positive (implied > realised)	contango (VIX < VIX3M)	**short**, full size
negative	contango	**short**, half size
negative	backwardation (VIX > VIX3M)	**long**, full size
positive	backwardation	**cash**

The fourth row is the interesting one: when the curve is inverted the market is paying up for near-term protection, so the premium seller stands down. Being long requires both a stressed curve *and* implied volatility that has stopped exceeding realised — the crash has already happened.



2020: VIX (red) against VIX3M (blue). Green shading is the long-vol regime, amber the cash regime. The lower panel is the resulting position as a fraction of NAV — note how often it flips in March, and how the largest drawdown of the whole backtest happens immediately *after* the long window, when implied vol collapses and the long leg gives the money back.

2 · The rules, stated exactly

Everything below is required to reproduce the results. Where the source does not specify a quantity, the choice made here is marked CHOICE .

INPUTS, ONE VALUE PER TRADING SESSION

SERIES	USE	SOURCE USED HERE
VIX (30-day implied vol index)	implied volatility, term-structure short leg	Cboe ^VIX
VIX3M (3-month implied vol index)	term-structure long leg	Cboe ^VIX3M
S&P 500 close	realised volatility for VRP	index close
VIXY or VXX	the traded instrument (ETN/ETF version)	split- and dividend-adjusted close
VIX futures settlements	the traded instrument (futures version)	Cboe CFE contract files

SIGNAL 1 — VOLATILITY RISK PREMIUM

```
realised vol RV20 = 100 × stdev( ln( SPX_t / SPX_t-1 ) for the last 20 sessions ) × √252
VRP = VIX − RV20 // both in volatility points
```

Twenty sessions is a CHOICE : the source says only "what recent price action suggests it should be". Ten and sixty session windows are tested later and the conclusion does not depend on the choice. Positive VRP means implied is above realised.

SIGNAL 2 — TERM STRUCTURE

```
contango = VIX < VIX3M // normal, calm market → short-friendly
inverted = VIX ≥ VIX3M // stressed → the only route to long
```

WHAT THE SOURCE CALLS PROPRIETARY

The exact moving-average pair in the ETF sleeve (out of scope here) and the precise VRP estimator. Every other rule — the four-row table, the VIX/100 sizing, the 40% cap, the Friday cadence, the 3:30pm signal, the cost assumptions — is stated by the source and used verbatim.

2 · The rules, stated exactly (continued)

POSITION SIZE

```
base = min( VIX / 100 , 0.40 ) // VIX 15 → 15% of NAV, cap 40%
size = base // short full / long full
size = 0.5 × base // short-half row of the table
weight = −size (short) | +size (long) | 0 (cash)
```

Scaling with the VIX level is the source's own rule: volatility is mean-reverting, so a higher VIX both raises the expected reversion and the premium being paid. The 40% cap is a hard limit on the sleeve, and it is enforced **daily** — a short that drifts above 40% because the market moved is trimmed the same session, not at the next rebalance.

WHEN TO TRADE

- Once a week, on the **last trading session of the week** (Friday, or the session before a holiday), at the close.
- Immediately on any session where the **regime changes** from the previous session, at that close.
- Fills are **market-on-close**: one official auction price, no slippage modelled.
- Signals are read from the **same session's** data — the source specifies a 3:30pm read with a 4:00pm fill. A previous-close variant is reported as the causal bound.

COSTS

ITEM	ETN/ETF VERSION	FUTURES VERSION
Commission	$0.005 per share traded	≈$1.50 per contract round turn (<1bp)
Borrow	**3.5% per year on short notional**, accrued daily	none — futures are unfunded
Management fee	0.85%/yr inside the ETF price (collected by the short)	none
Cash interest	0% (conservative; a margin account earns it)	0% (same convention)
Slippage	0 — market-on-close	0 — settlement

3 · How to reproduce it — step by step

No code is needed; the procedure is arithmetic on daily closes. Follow it in order and you will land on the numbers in section 4.

PREPARATION

1. Collect daily closes for VIX, VIX3M, the S&P 500 and your instrument, **from 2011-01-04**.
2. Use **adjusted** instrument prices. Both tradable VIX ETNs have done multiple reverse splits — VIXY on 2013-06-10 (1:5), 2016-07-25 (1:5), 2017-07-17 (1:4), 2021-05-26 (1:4), 2023-06-23 (1:5) and 2024-11-07 (1:4); VXX on 2021-04-23, 2023-03-07 and 2024-07-24 (1:4 each). Raw prices show −75% to −80% one-day "crashes" on those dates that never happened to a holder.
3. Start with NAV = 1.00 and zero shares. The first 20 sessions are needed for RV20, so the first investable day is the 21st.

THE DAILY LOOP

1. **Mark to market.** NAV gains `shares × (close_t − close_t−1)`. If short, subtract `(3.5%/252) × |shares| × close_t−1`.
2. **Update the regime** from today's closes: compute RV20, VRP = VIX − RV20, compare VIX with VIX3M, and read the four-row decision table.
3. **Decide whether to trade.** Trade if today is the week's last session, or if the regime differs from yesterday's. Otherwise keep the existing share count.
4. **Trade at the close.** Target shares = `weight × NAV / close_t`. Charge $0.005 per share of the difference. The target weight comes from the sizing formula, using today's VIX.
5. **Apply the cap.** If the position's absolute value now exceeds 40% of NAV, trade back to exactly 40% and charge the commission.
6. **Record** NAV and the position weight. Repeat.

> **THREE DETAILS THAT DECIDE WHETHER YOU LAND ON THE SAME NUMBERS**
>
> **(a)** Weights are fractions of NAV, not of the sleeve. **(b)** The regime is recomputed every session but only *acted on* on trade days — that is what makes the position drift between Fridays and why some days are trimmed by the cap. **(c)** RV20 uses log returns of the index close, annualised with 252 sessions; a 21-day count or simple returns shifts the results by a few tenths of a point, not the conclusion.

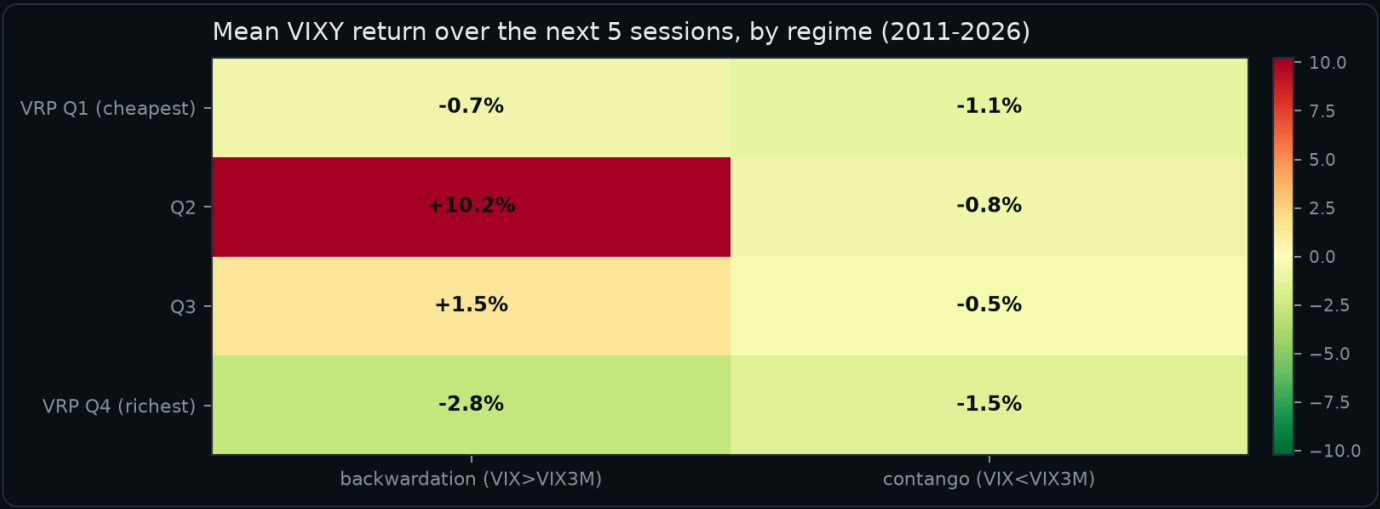
FOR THE FUTURES VERSION

1. Hold the **nearest monthly VIX future** (third-Wednesday expiry), sized the same way: target notional = weight × NAV, contracts = notional / (price × $1,000 per point).
2. Roll on the **session before expiry**: close the expiring contract, open the next one. The roll itself is P&L-neutral; what you collect is each contract's own price change plus the fact that the new contract is priced higher in contango.
3. Charge ≈$1.50 per contract round turn and **no borrow**. Margin is 5-10% of notional, so a 40% notional position ties up 2-4% of NAV in collateral rather than 40% of cash.

> **CHECK YOUR IMPLEMENTATION**
>
> Before comparing returns, compare the **regime mix**: over 2011-2026 roughly 77% of sessions should read *short, full size*, 12% *short, half*, 3% *long* and 8% *cash*. If your mix is far from that, the sign of the VRP test or the direction of the curve comparison is inverted — fix that before looking at a single P&L number.

4 · Why it works, measured

Bucket every session 2011-2026 by its VRP quartile and by the shape of the curve, then look at what the short-volatility instrument actually did over the next five sessions. If the logic is sound the top-left corner should be the only reliably profitable place to be short, and the bottom-right the only place to be long.



Mean five-day VIXY return: **negative** where VRP is rich and the curve is in contango (short pays), **positive** where VRP is cheap and the curve is inverted (long pays). The gradient is monotone in both directions, which is exactly what the decision table encodes. That is the whole edge — no forecast, just a conditional premium.

- **Short pays only where VRP is rich.** The top row of the map is the strategy's bread and butter: implied above realised, curve upward sloping.
- **Long pays only where the curve is inverted and VRP has turned negative** — the crash has already happened and implied vol is still catching up with realised. That is 2-3% of sessions, and it is the only thing that stands between this book and a 47% drawdown.
- **Everything else is cash.** Rich VRP with an inverted curve means the market is paying up for protection you would be selling — the worst combination available.

TYPICAL REGIME SPLIT, 2011-2026

short, full	76.8%
short, half	12.4%
long	3.3%
cash	7.6%

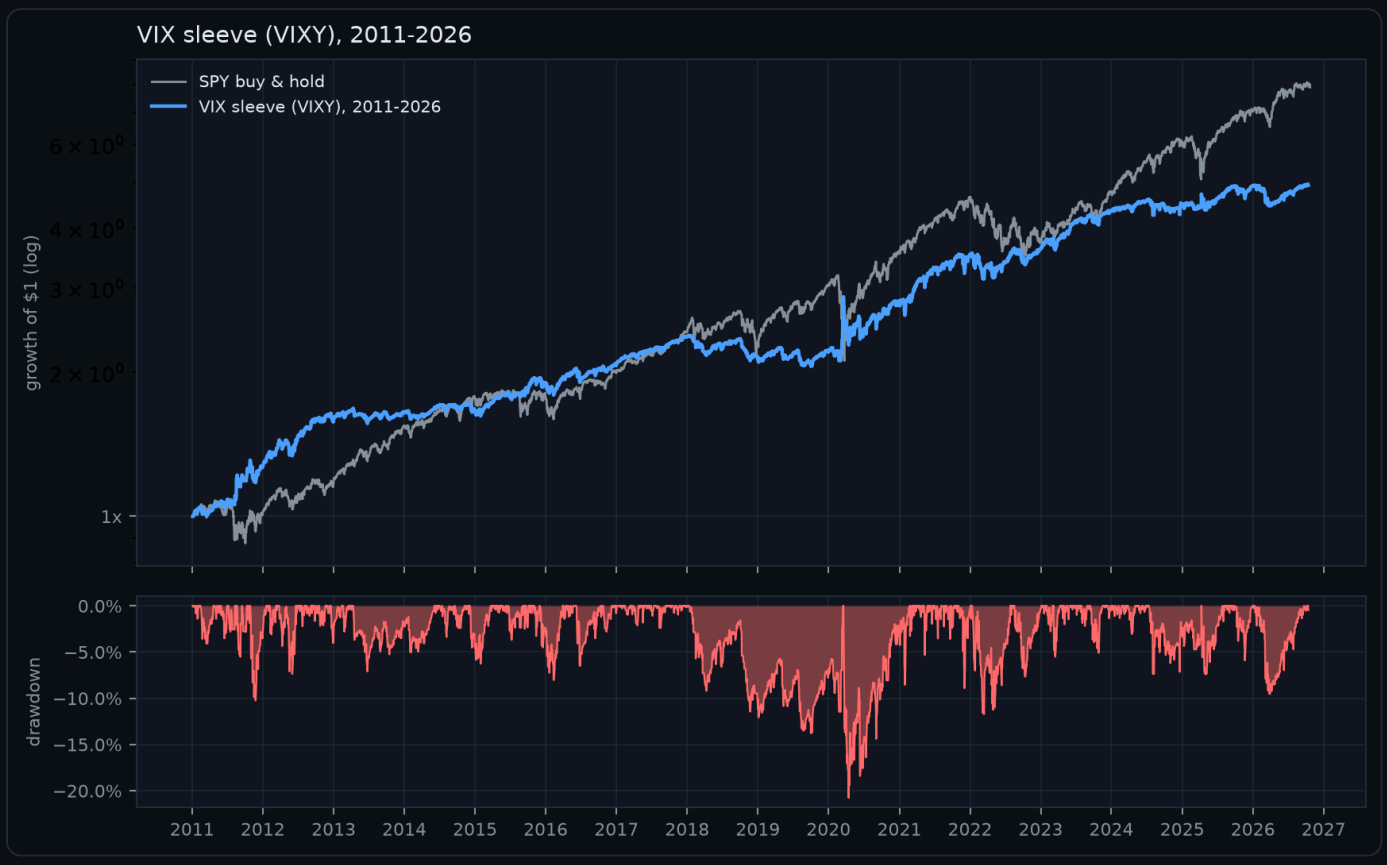
The research publishes 91.40% short / 1.99% long / 6.61% cash in total. The replication lands on 92.4 / 2.2 / 5.4 — the strongest available evidence that the rule was decoded correctly, since this statistic cannot be gamed by tuning returns.

WHAT THE SIGNALS ARE WORTH

Always short, no signal: 7.9% CAGR with a -47.1% drawdown. With the signals: 10.6% and -20.8%. The signal is not adding return so much as **removing the tail** — it more than halves the worst drawdown and turns a 0.56 Sharpe into 0.85.

5 · Results — the ETN/ETF version

Instrument: **VIXY**, ProShares VIX Short-Term Futures ETF, 2011-01-04 → 2026-10-14. VXX is the instrument the source names, but the iPath VXX ETN was redeemed in 2018-01 and no free continuous series covers the earlier window; VIXY holds the same front-two-month VIX futures basket and has traded since the start of the research period. On the overlapping 2018+ window the two differ by 0.24pp per year.



WINDOW	CAGR	VOL	SHARPE	MAX DD	CALMAR
Full 2011-2026	10.6%	12.9%	0.85	-20.8%	0.51
2011-2021	12.1%	13.3%	0.93	-20.8%	0.58
2022-2026	7.1%	11.8%	0.65	-11.7%	0.61
SPY, same window	14.0%	17.0%	0.86	-33.7%	0.41

YEAR BY YEAR

YEAR	RETURN	YEAR	RETURN	YEAR	RETURN	YEAR	RETURN
2011	26.4%	2015	14.8%	2019	3.6%	2023	20.4%
2012	24.7%	2016	8.9%	2020	27.1%	2024	2.3%
2013	4.6%	2017	14.8%	2021	25.0%	2025	9.9%
2014	0.2%	2018	-9.7%	2022	3.0%	2026	0.3%

The shape is the strategy: many boring positive years in the low-to-mid teens, one genuinely bad year (2018, −9.7%), and a 2020 that pays for everything else. Note that 2020's +27% and the backtest's worst drawdown (−20.8%) happen in the same year, three weeks apart.

6 · What drives the ETF result

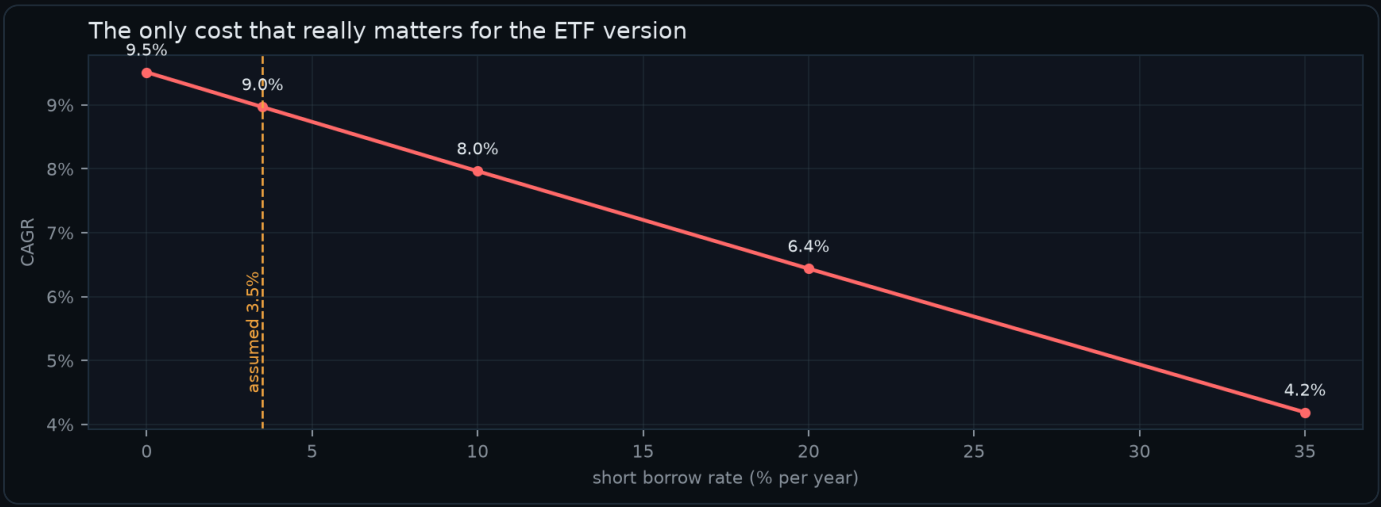
EACH PIECE, ON ITS OWN

VARIANT	CAGR	VOL	SHARPE	MAX DD	CALMAR	WHAT IT ISOLATES
Full rule (base case)	10.6%	12.9%	0.85	-20.8%	0.51	
Short-only	9.4%	10.9%	0.88	-11.7%	0.80	drop the long leg
Fixed 10% size	4.5%	5.9%	0.77	-9.0%	0.50	no VIX scaling
VRP leg only	8.6%	13.7%	0.67	-28.7%	0.30	no term structure
Term structure only	10.8%	16.0%	0.72	-22.3%	0.48	no VRP filter
Always short	7.9%	15.9%	0.56	-47.1%	0.17	no signals at all

Timing is the strategy. Delay the signal by one session — read yesterday's closes instead of today's — and the same rules give **5.7%** CAGR with a **-30.2%** drawdown instead of 10.6% and -20.8%. The edge decays inside 24 hours; it is a same-day volatility read, not a weekly set-and-forget. Anyone implementing it needs a mid-afternoon VIX/VIX3M snapshot and a closing-auction order, which is precisely what the source's platform sells.

Costs are secondary. Removing the commission and the borrow entirely lifts CAGR from 10.6% to 11.3% — about 0.7pp, almost all of it borrow.

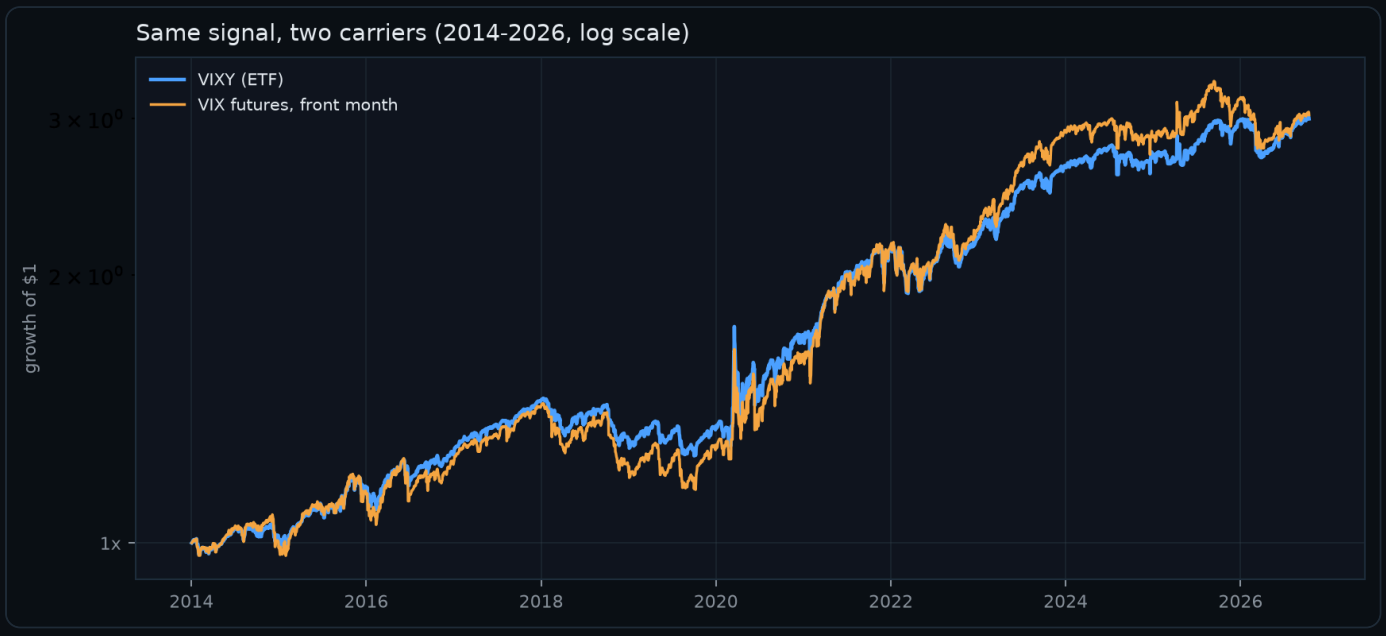
Read the table, not just the ranking: the VIX-scaled book compounds faster than the fixed-size one because it leans in exactly when mean reversion is largest; the short-only book gives up the 2020 spike but avoids most of the drawdown. Neither ablation improves the full rule's risk-adjusted result, which is why all three rules are kept.



The same book at different borrow assumptions. 3.5% is the source's own assumption and a calm-market average; short-volatility ETNs become expensive — or impossible — to borrow exactly when volatility spikes. That asymmetry is invisible in a backtest and is the main practical argument for the futures implementation in section 8.

7 · Results — the VIX futures version

Same signal table, same sizing, same 40% cap, same rebalance rule. The instrument is the **front-month VIX future**, rolled the session before expiry, rebuilt from Cboe's own contract files (154 monthly contracts, 2014-01-02 → 2026-10-22 — the archive begins in 2013).



CARRIER	CAGR	VOL	SHARPE	MAX DD	CALMAR
VIXY, 3.5% borrow (published)	9.0%	12.9%	0.73	-20.8%	0.43
VIXY, zero borrow	9.5%	12.9%	0.77	-20.8%	0.46
VIX futures, front month	9.1%	15.4%	0.64	-20.7%	0.44
VIX futures, always short	8.3%	19.2%	0.51	-47.2%	0.18
Constant 30-day basket (diagnostic)	7.7%	13.5%	0.61	-21.1%	0.36

WHY THE TWO CARRIERS DIVERGE

	VIXY (ETF)	FRONT-MONTH FUTURES
Long-only decay, 2014-2026	−45.95% per year	**−54.02%** per year
Long-only volatility	70.4%	**87.3%**
Sharpe of the raw short	0.54	0.48
2018 (volmageddon)	−9.7%	**−14.4%**
2020 (COVID)	+27.1%	+27.0%
2021 / 2023 (steep contango)	+25.0% / +20.4%	**+32.4% / +29.1%**
2024 (calm, bad rolls)	+2.3%	−1.7%

The front month decays about **8 points per year faster** than the ETF — that is the extra carry a short collects — but it arrives with **17 points more volatility**. Identical CAGR, more risk: the futures version is a levered version of the same bet, not a better one. The one exception is 2018, where a single-day spike in the front of the curve costs the futures book almost five extra points.

8 · Which wrapper should you use?

	ETN / ETF (VIXY, VXX)	VIX FUTURES
Expense ratio	0.85%/yr — **collected** by a short seller	none (and nothing to collect)
Short borrow	3.5% assumed; 10-35% or unavailable in a squeeze	**none**
Commission	$0.005/share	≈$1.50/contract round turn
Capital per $1 of notional	100% cash, or 50% margin + borrow	**5-10% margin**
Contract granularity	one share, ≈$18	≈$20,000 of notional at VIX 20
Settlement	4:00pm closing auction — matches the source's MOC rule	4:15pm settle, no auction
Issuer risk	ETN redemption / creation halts (VXX, Feb 2018)	exchange-cleared
Roll control	the fund rolls for you, daily, constant maturity	you roll monthly, sawtooth maturity

VERDICT

On expected return the two are a wash — 9.0% versus 9.1% CAGR over the shared 2014-2026 window. Futures win on robustness and capital efficiency, not on return: no borrow market to disappear on you, no issuer to fail, and 5-10× less capital per unit of notional (worth roughly 0.2-0.3pp per year of T-bill interest on the freed collateral at current rates). The ETF wins on simplicity, granularity, and the fact that its 4:00pm closing auction is exactly the fill convention the strategy was designed around.

YEAR	VIXY	FRONT MONTH
2014	0.2%	-1.2%
2015	14.8%	15.3%
2016	8.9%	7.6%
2017	14.8%	15.9%
2018	-9.7%	-14.4%
2019	3.6%	5.5%
2020	27.1%	27.0%
2021	25.0%	32.4%
2022	3.0%	4.4%
2023	20.4%	29.1%
2024	2.3%	-1.7%
2025	9.9%	10.5%
2026	0.3%	-4.1%

9 · Caveats, and how this could be wrong

RULES OF THUMB, RESTATED

- **Small account, or borrow available at ≤3.5%:** use the ETF. Simpler, finer sizing, marginally better risk-adjusted (0.73 vs 0.64 Sharpe).
- **More than roughly $1-2M of notional, or you cannot tolerate a borrow squeeze:** use futures. You give up ~0.1pp of fee collection and gain immunity to the single worst operational risk in the strategy.
- **Never** run the futures version without checking that a 40% notional position cannot be liquidated by a single-day variation-margin call. In March 2020 the front month moved more than 30% in a session.
- Either way, the sizing cap is a **notional** cap. In futures, 40% of NAV notional is 4-8% of NAV in margin — do not mistake that for a smaller position.

- **The 3:30pm read is approximated by the close.** The source reads VIX at 3:30pm and fills at 4:00pm. Daily bars cannot see 3:30pm, so the base case reads the session close and fills at that same close. VIX settles at 4:15pm, so the last 30 minutes of the VIX session are technically visible — a mild look-ahead. The previous-close variant (5.7% CAGR, -30.2% drawdown) is the fully causal bound. **The truth is between the two, and the gap is large: this is the single most important number in the document.**
- **VIXY is not VXX.** The named instrument was redeemed in 2018-01. VIXY tracks the same basket; on the 2018+ overlap the rule set differs by 0.24pp per year.
- **VIX3M is not the futures curve.** The source illustrates the term structure with the actual futures curve; VIX3M is the free daily stand-in. A real curve would sharpen the long-vol filter.
- **The futures archive starts in 2013-2014.** The futures comparison therefore cannot see 2011-2013, when the curve was also steeply in contango.
- **No cash interest is credited** anywhere. The source describes a margin account that earns it. On the sleeve alone that is worth roughly 0.3-0.5pp per year.
- **Borrow is modelled at a constant 3.5%.** Reality is state-dependent, and worst exactly when the position is largest.
- **No slippage.** Defensible for market-on-close in liquid ETFs at retail size; less so in VIX futures at size near expiry.

PUBLISHED VS REPLICATED FINGERPRINTS

CHECK	PUBLISHED	REPLICATION
Time short / long / cash	91.40% / 1.99% / 6.61%	**92.41% / 2.19% / 5.39%**
Worst year	2018, about −15%	**−13.6%**
2022	effectively flat	**−2.6%**
Best year	2020	**+40.2%** (portfolio level)
Profitable years	13 of 15	**13 of 15**

WHAT WOULD CHANGE THE VERDICT

A real 3:30pm dataset. If a reader can source intraday VIX/VIX3M snapshots, the honest number moves toward the top of the bracket rather than the middle — this is the one input that would settle the biggest open question in the document. **A longer futures archive** (2011-2013) would test whether the futures-vs-ETF conclusion survives a second steep-contango regime. **Real borrow data** for VIXY/VXX would replace the flat 3.5% assumption with the actual, state-dependent cost.

Appendix · Every book, headline numbers

2011-01-04 → 2026-10-14 for the ETF books; 2014-01-02 → 2026-10-22 for the futures books. These are the figures to check your own implementation against.

BOOK	CAGR	VOL	SHARPE	MAX DD	CALMAR
Base case — VIXY, same-session signal	10.6%	12.9%	0.85	-20.8%	0.51
Previous-close signal (causal bound)	5.7%	12.4%	0.51	-30.2%	0.19
Zero commission and zero borrow	11.3%	12.9%	0.90	-20.8%	0.54
Short-only (no long-vol leg)	9.4%	10.9%	0.88	-11.7%	0.80
Fixed 10% size (no VIX scaling)	4.5%	5.9%	0.77	-9.0%	0.50
VRP leg only	8.6%	13.7%	0.67	-28.7%	0.30
Term-structure leg only	10.8%	16.0%	0.72	-22.3%	0.48
Always short VIXY (no signal)	7.9%	15.9%	0.56	-47.1%	0.17
SPY buy & hold	14.0%	17.0%	0.86	-33.7%	0.41
Futures — front month, 2014+	9.1%	15.4%	0.64	-20.7%	0.44
Futures — constant 30-day basket	7.7%	13.5%	0.61	-21.1%	0.36
Futures — always short front month	8.3%	19.2%	0.51	-47.2%	0.18
VIXY on the futures window, 2014+	9.0%	12.9%	0.73	-20.8%	0.43
VIXY on the futures window, zero borrow	9.5%	12.9%	0.77	-20.8%	0.46

The portfolio-level figures quoted above come from the combined book in the project's main report; the sleeve alone makes 10.6% a year with a -20.8% drawdown.

DATA USED

SERIES	SOURCE	WINDOW
VIX, VIX3M, VIX9D index closes	Cboe index history	2004 → 2026-10
S&P 500 daily close	index history	1950 → 2026-10
VIXY, VXX (split- and dividend-adjusted)	Yahoo Finance daily bars	2011-01-04 / 2018-01-25 → 2026-10
VIX futures settlements, 154 monthly contracts	Cboe CFE per-contract CSV archive	2013-09 → 2026-10
VIX futures, continuous front-month	built here: nearest expiry, rolled the session before expiry	2014-01-02 → 2026-10-22